



Barker
College

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Student Number

2021

YEAR 12
ASSESSMENT TASK 3

Mathematics Extension 2

Staff Involved:

BHC* ARM

40 copies

8:30AM
WEDNESDAY 2 JUNE

TOPICS COVERED

- Integration
- Vectors

General

Instructions:

- Working time – 55 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- Write all answers in the spaces provided
- Show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale
- Extra working space is provided at the end of the paper

Total marks:

48

Question 1 (11 marks)

Marks

(a) Find the vector $\underline{v}$ parallel to $\underline{u} = 2\underline{i} + 3\underline{j} + 4\underline{k}$ that has a magnitude of 5.

2

(b) Find the angle between the vectors _____ and _____, rounded to the nearest degree.

3

(c) Consider the points $A(1, 3, 2)$ and $B(2, 4, -1)$.

(i) Find a vector equation of the line passing through A and B .

1

(ii) Prove that the point $C(5, 7, -10)$ lies on the line passing through A and B .

1

(iii) Using (i), find a vector equation for the interval from A to B .

1

(d) The points $A(5\frac{2}{3}, 3, 4)$, $B(5, 6, z)$ and $C(8, 9, 10)$ form a right-angled triangle, right-angled at B . Find the possible values of z .

3

Question 2 (12 marks)

Marks

(a) Find $\int \frac{2x+1}{x^2+4x+5} dx$.

3

(b) Use the substitution $x = 2 \sin \theta$ to evaluate

3

Question 3 (9 marks)

Marks

- (a) Evaluate $\int_0^1 \frac{x^2+3}{(x^2+1)(x^2+2)} dx$ using partial fractions, correct to 2 decimal places. **3**

Question 4 (16 marks)**Marks**

- (a) Find the point of intersection of the lines

3

$$\underline{r} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{and} \quad \underline{q} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 8 \\ 7 \end{pmatrix}.$$

- (b) The point $B(x, y, z)$ is on the interval AC and is twice as far from $A(-1, -3, -5)$ as it is from $C(5, 6, 7)$. Use vectors to find the coordinates of B .

3

(d) (i) If $I_n = \int_0^1 (x^2 - 1)^n dx$ for $n \geq 0$, show that $I_n = -\frac{2}{n+1} I_{n-1}$ for $n \geq 1$. **3**

(ii) Hence, evaluate I_3 . **1**

Question 1 (11 marks)

Marks

- (a) Find the vector $\underline{v}$ parallel to $\underline{u} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ that has a magnitude of 5.

2

$$\text{Now } |\underline{u}| = \sqrt{4+9+16} = \sqrt{29}$$

$$\text{so } \underline{u} = \begin{pmatrix} \frac{2}{\sqrt{29}} \\ \frac{3}{\sqrt{29}} \\ \frac{4}{\sqrt{29}} \end{pmatrix}$$

$$\therefore \underline{v} = \frac{1}{\sqrt{29}} \begin{pmatrix} 10 \\ 15 \\ 20 \end{pmatrix}$$

- (b) Find the angle between the vectors $\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}$, rounded to the nearest degree.

3

$$\cos \theta = \frac{\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}}{\sqrt{29} \times \sqrt{8}} = \frac{2}{\sqrt{29} \times \sqrt{8}}$$

$$\therefore \cos \theta = 0.131306$$

$$\therefore \theta = 82^\circ$$

- (c) Consider the points $A(1, 3, 2)$ and $B(2, 4, -1)$.

- (i) Find a vector equation of the line passing through A and B .

1

$$\underline{AB} = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}$$

$$\therefore \underline{r} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}$$

$$\text{or } \underline{r} = \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}$$

$$\text{or } \underline{r} = \begin{pmatrix} 2 \\ 7 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -3 \\ 3 \end{pmatrix} \quad \text{or } \underline{r} = \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}$$

- (ii) Prove that the point $C(5, 7, -10)$ lies on the line passing through A and B .

1

$$\text{Does } \begin{pmatrix} 5 \\ 7 \\ -10 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} ?$$

Yes, when $\lambda = 4$

(or equivalent)

- (iii) Using (i), find a vector equation for the interval from A to B .

1

The interval from A to B is

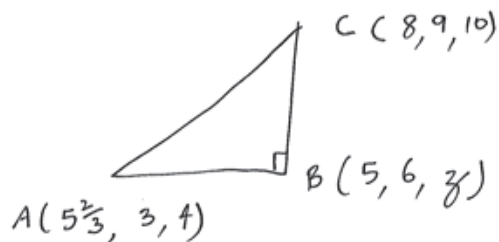
$$\underline{r} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} \quad \text{for } 0 \leq \lambda \leq 1$$

$$\text{or } \underline{r} = \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix} \quad \text{for } -1 \leq \lambda \leq 0$$

$$\text{or } \underline{r} = \begin{pmatrix} 2 \\ 7 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -3 \\ 3 \end{pmatrix} \quad \text{for } -1 \leq \lambda \leq 0$$

$$\text{or } \underline{r} = \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix} \quad \text{for } 0 \leq \lambda \leq 1$$

- (d) The points $A(5\frac{2}{3}, 3, 4)$, $B(5, 6, z)$ and $C(8, 9, 10)$ form a right-angled triangle, right-angled at B . Find the possible values of z .



Now $\vec{BC} \cdot \vec{AB} = 0$, so

$$\begin{pmatrix} 3 \\ 3 \\ 10-z \end{pmatrix} \cdot \begin{pmatrix} -\frac{2}{3} \\ 3 \\ z-4 \end{pmatrix} = 0$$

$$\therefore -2 + 9 + (10-z)(z-4) = 0$$

$$7 + 10z - 40 - z^2 + 4z = 0$$

$$-z^2 + 14z - 33 = 0$$

$$z^2 - 14z + 33 = 0$$

$$(z-3)(z-11) = 0$$

$$\therefore z = 3 \text{ or } 11$$

Question 2 (12 marks)

Marks

- (a) Find $\int \frac{2x+1}{x^2+4x+5} dx$.

3

$$= \int \frac{2x+4-3}{x^2+4x+5} dx$$

$$= \int \frac{2x+4}{x^2+4x+5} dx - \int \frac{3}{x^2+4x+5} dx$$

$$= \int \frac{2x+4}{x^2+4x+5} dx - \int \frac{3}{1+(x+2)^2} dx$$

$$= \ln|x^2+4x+5| - 3 \tan^{-1}(x+2) + C$$

- (b) Use the substitution $x = 2 \sin \theta$ to evaluate $\int_0^{\sqrt{2}} \frac{dx}{\sqrt{(4-x^2)^3}}$.

3

$$\text{let } x = 2 \sin \theta$$

$$dx = 2 \cos \theta d\theta$$

$$\text{when } x = \sqrt{2}, \theta = \frac{\pi}{4}$$

$$\text{when } x = 0, \theta = 0$$

$$\therefore I = \int_0^{\frac{\pi}{4}} \frac{2 \cos \theta d\theta}{\sqrt{(4-4 \sin^2 \theta)^3}}$$

$$= \int_0^{\frac{\pi}{4}} \frac{2 \cos \theta}{8 \sqrt{\cos^6 \theta}} d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{4} \sec^2 \theta d\theta$$

$$= \frac{1}{4} [\tan \theta]_0^{\frac{\pi}{4}} = \frac{1}{4}$$

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Student Number

Marks

(c) Find $\int x^2 \sin x dx$.

3

$$\begin{aligned}
 &= -\cos x \cdot x^2 - \int (-\cos x) \cdot 2x dx \\
 &= -x^2 \cos x + 2 \int x \cos x dx \\
 &= -x^2 \cos x + 2 \left[\sin x \cdot x - \int \sin x dx \right] \\
 &= -x^2 \cos x + 2x \sin x - 2 \int \sin x dx \\
 &= -x^2 \cos x + 2x \sin x + 2 \cos x + C
 \end{aligned}$$

(d) Find $\int (\ln x)^2 dx$.

3

$$\begin{aligned}
 &= x (\ln x)^2 - \int x \cdot 2 (\ln x)' \cdot \frac{1}{x} dx \\
 &= x (\ln x)^2 - 2 \int \ln x dx \\
 &= x (\ln x)^2 - 2 \left[x \ln x - \int x \left(\frac{1}{x} \right) dx \right] \\
 &= x (\ln x)^2 - 2x \ln x + 2 \int dx \\
 &= x (\ln x)^2 - 2x \ln x + 2x + C
 \end{aligned}$$

Question 3 (9 marks)

(a) Evaluate $\int_0^1 \frac{x^2+3}{(x^2+1)(x^2+2)} dx$ using partial fractions, correct to 2 decimal places.

3

$$\text{let } \frac{x^2+3}{(x^2+1)(x^2+2)} = \frac{A}{x^2+1} + \frac{B}{x^2+2}$$

$$x^2+3 = A(x^2+2) + B(x^2+1)$$

$$\begin{aligned}
 \text{let } x=0, & \quad 3 = 2A + B \\
 \text{let } x=1, & \quad 4 = 3A + 2B \\
 & \quad 6 = 4A + 2B \\
 \hline
 & \quad 2 = A \\
 & \quad B = -1
 \end{aligned}$$

$$\begin{aligned}
 \therefore I &= \int_0^1 \frac{2}{1+x^2} + \frac{-1}{2+x^2} dx \\
 &= \left[2 \tan^{-1} x \right]_0^1 - \left[\frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} \right]_0^1 \\
 &= 2 \left(\frac{\pi}{4} - 0 \right) - \frac{1}{\sqrt{2}} \left(\tan^{-1} \frac{1}{\sqrt{2}} \right) \\
 &= \frac{\pi}{2} - \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{1}{\sqrt{2}} \right) \\
 &= 1.1355... \\
 &= 1.14 \text{ (2dp)}
 \end{aligned}$$

- (b) Let $u = \sqrt{x}$ to evaluate $\int_1^9 e^{\sqrt{x}} dx$.

3

$$\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$2u du = dx$$

$$\text{When } x=9, u=3$$

$$\text{When } x=1, u=1$$

$$I = \int_1^9 e^u \cdot 2u du$$

$$= 2 \int_1^9 u e^u du$$

$$= 2 \left[[e^u \cdot u]_1^9 - \int_1^9 e^u (1) du \right]$$

$$= 2 \left[3e^3 - e - [e^u]_1^9 \right]$$

$$= 2 \left[3e^3 - e - (e^3 - e) \right]$$

$$= 2(2e^3) = 4e^3$$

- (c) Use an appropriate substitution to find $\int \frac{dx}{\sqrt{x}\sqrt{1-x}}$.

3

Marks will only be awarded for solutions using a substitution.

$$\text{Let } u = x$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

$$I = \int \frac{2u du}{u \cdot \sqrt{1-u^2}}$$

$$= 2 \int \frac{du}{\sqrt{1-u^2}}$$

$$= 2 \sin^{-1}(u)$$

$$= 2 \sin^{-1}(\sqrt{x}) + C$$

$$\text{OR Let } x = \sin^2 \theta$$

$$dx = 2 \sin \theta \cos \theta d\theta$$

$$I = \int \frac{2 \sin \theta \cos \theta d\theta}{\sin \theta \cdot \cos \theta}$$

$$= \int 2 d\theta$$

$$= 2\theta$$

$$\text{but } \sin \theta = \sqrt{x} \text{ so}$$

$$\theta = \sin^{-1}(\sqrt{x})$$

$$\text{Hence } I = 2 \sin^{-1}(\sqrt{x}) + C$$

Question 4 (16 marks)

Marks

- (a) Find the point of intersection of the lines

3

$$r = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{and} \quad q = \begin{pmatrix} 3 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 8 \\ 7 \end{pmatrix}$$

$$\begin{pmatrix} 1+2\lambda \\ 1+3\lambda \end{pmatrix} = \begin{pmatrix} 3+8\mu \\ -1+7\mu \end{pmatrix}$$

$$\begin{aligned} 2\lambda - 8\mu &= 2 \\ 3\lambda - 7\mu &= -2 \end{aligned} \Rightarrow \begin{aligned} \lambda - 4\mu &= 1 \\ 3\lambda - 7\mu &= -2 \\ 3\lambda - 12\mu &= 3 \end{aligned}$$

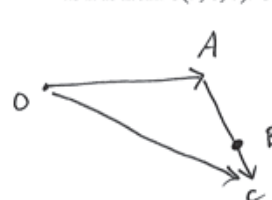
$$\therefore 5\mu = -5$$

$$\therefore \mu = -1 \text{ (OR } \lambda = -3)$$

$$\therefore \text{Point of intersection is } \begin{pmatrix} -5 \\ -8 \end{pmatrix}$$

- (b) The point $B(x, y, z)$ is on the interval AC and is twice as far from $A(-1, -3, -5)$ as it is from $C(5, 6, 7)$. Use vectors to find the coordinates of B .

3



$$\vec{AC} = \vec{OC} - \vec{OA} = \begin{pmatrix} 6 \\ 9 \\ 12 \end{pmatrix}$$

$$\vec{OB} = \vec{OA} + \frac{2}{3} \vec{AC}$$

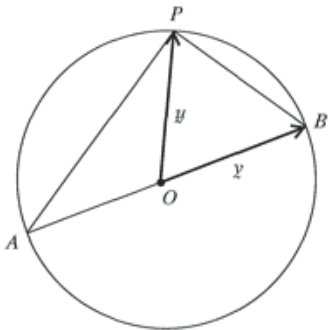
$$= \begin{pmatrix} -1 \\ -3 \\ -5 \end{pmatrix} + \frac{2}{3} \begin{pmatrix} 6 \\ 9 \\ 12 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ -3 \\ -5 \end{pmatrix} + \begin{pmatrix} 4 \\ 6 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

- (c) AB is the diameter of a circle, centre O . Let $\vec{OB} = \mathbf{v}$ and $\vec{OP} = \mathbf{u}$.

Use vectors to prove that $\angle APB = 90^\circ$.



$$\text{Now } \vec{OA} = -\mathbf{v}$$

$$\vec{AP} = \mathbf{u} - (-\mathbf{v}) = \mathbf{u} + \mathbf{v}$$

$$\vec{PB} = \mathbf{v} - \mathbf{u}$$

$$\vec{AP} \cdot \vec{PB} = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{v} - \mathbf{u})$$

$$= \mathbf{u} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{v}$$

$$= \mathbf{v} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{u}$$

$$= |\mathbf{v}|^2 - |\mathbf{u}|^2$$

$$= 0 \quad (\text{since } |\mathbf{v}| = |\mathbf{u}| = \text{radius})$$

$$\therefore \angle APB = 90^\circ$$

3

- (d) (i) If $I_n = \int_0^1 (x^2 - 1)^n dx$ for $n \geq 0$, show that $I_n = \frac{-2n}{2n+1} I_{n-1}$ for $n \geq 1$.

3

$$\begin{aligned} I_n &= \int_0^1 (x^2 - 1)^n dx \\ &= \left[x (x^2 - 1)^n \right]_0^1 - \int_0^1 x \cdot n (x^2 - 1)^{n-1} (2x) dx \\ &= 0 - 2n \int_0^1 x^2 (x^2 - 1)^{n-1} dx \\ &= -2n \int_0^1 (x^2 - 1)(x^2 - 1)^{n-1} + (x^2 - 1)^{n-1} dx \\ &= -2n \int_0^1 (x^2 - 1)^n + (x^2 - 1)^{n-1} dx \\ &= -2n I_n - 2n I_{n-1} \end{aligned}$$

$$2n I_n + I_n = -2n I_{n-1}$$

$$I_n(2n+1) = -2n I_{n-1} \quad \therefore I_n = \frac{-2n}{2n+1} I_{n-1}$$

- (ii) Hence, evaluate I_3 .

1

$$I_3 = -\frac{6}{7} I_2$$

$$I_2 = -\frac{4}{5} I_1$$

$$I_1 = -\frac{2}{3} I_0$$

$$\begin{aligned} I_0 &= \int_0^1 (x^2 - 1)^0 dx \\ &= [x]_0^1 \\ &= 1 \end{aligned}$$

$$\therefore I_1 = -\frac{2}{3}$$

$$\therefore I_2 = -\frac{4}{5} \times -\frac{2}{3}$$

$$= \frac{8}{15}$$

$$\therefore I_3 = -\frac{6}{7} \times \frac{8}{15}$$

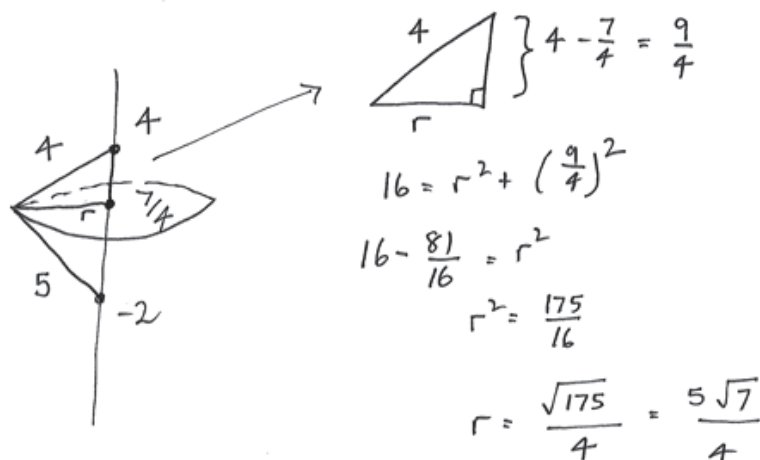
$$= -\frac{48}{105} = -\frac{16}{35}$$

- (c) The spheres $(x+2)^2 + (y+3)^2 + (z-4)^2 = 16$ and $(x+2)^2 + (y+3)^2 + (z+2)^2 = 25$ intersect in a circle. What is the centre and radius of this circle?

3

$$\begin{aligned}(z+2)^2 - (z-4)^2 &= 25 - 16 \\ z^2 + 4z + 4 - (z^2 - 8z + 16) &= 9 \\ 12z - 12 &= 9 \\ z &= \frac{21}{12} = \frac{7}{4}\end{aligned}$$

$\therefore$ The centre is at $(-2, -3, \frac{7}{4})$



$$\begin{aligned}4 - \frac{7}{4} &= \frac{9}{4} \\ 16 &= r^2 + \left(\frac{9}{4}\right)^2 \\ 16 - \frac{81}{16} &= r^2 \\ r^2 &= \frac{175}{16} \\ r &= \frac{\sqrt{175}}{4} = \frac{5\sqrt{7}}{4}\end{aligned}$$

(OR substitute $z = \frac{7}{4}$ into either of the original equations)

End of Paper